

Lab 0: Python and Course Setup Guide

Information Retrieval Practical Laboratories | Windows x64 edition

The course runs with Python 3.10 or later and uses only the Python standard library. The included Python 3.13.15 installer is for Windows computers with a 64-bit Intel or AMD processor. Lab 0 lists all standard-library and included local imports in requirements.txt. No third-party pip packages are required.

Before you start

- Use Windows 10 or 11 x64, PowerShell or Windows Terminal, and enough space to extract the course ZIP and install Python. A code editor and PDF reader are useful but optional.
- For Windows on ARM, macOS, or Linux, get the matching installer from python.org/downloads.
- Keep each week's scripts with its docs and data folders. Lab programs load these files from beside the script.

1 Extract the course

Right-click **Information_Retrieval_Labs.zip** and choose **Extract All**. Open the extracted **Information_Retrieval_Labs** folder. Do not run scripts directly inside the ZIP.

2 Install Python

If Python 3.10 or later already works, you can skip this step. Otherwise:

- In Windows Settings, open **System > About > System type**. Confirm it says **64-bit operating system, x64-based processor**.

From the extracted course root, open the installer with this PowerShell command (or double-click the file in Lab_00_Setup):

```
.\Lab_00_Setup\python-3.13.15-amd64.exe
```

- Select **Add python.exe to PATH** if shown, then choose **Install Now**. The default setup includes pip, the standard library, and the Python launcher. You do not need optional free-threaded binaries or debug symbols.
- Finish setup and open a **new** PowerShell window.

3 Verify Python

Run these commands in the new PowerShell window:

```
py -3.13 --version
py -3.13 -c "import re, os, math, json; print('Course Python ready')"
```

Expected results: **Python 3.13.15** and **Course Python ready**. If you already had Python 3.10+, use **py -3**, **python**, or **python3** instead and check its version.

4 Check course requirements

From the extracted course root, run:

```
py -3.13 -m pip install --no-index -r .\Lab_00_Setup\requirements.txt
py -3.13 -c "import json, math, os, re, sys; print('Course requirements ready')"
```

The requirements file lists all five Python standard-library imports: **json**, **math**, **os**, **re**, **sys**. It also lists the seven local modules supplied with the labs: **evaluation**, **file_utils**, **indexing**, **preprocessing**, **ranking**, **retrieval**, **tfidf**. There is nothing for pip to download; --no-index works offline.

5 Run a laboratory

Open the extracted course folder in File Explorer. Click its address bar, type **powershell**, and press Enter. Then run:

```
py -3.13 .\Week_01_File_Manipulation\lab01_file_manipulation.py
py -3.13 .\Week_02_Regex_Text_Processing\lab02_regex.py
```

Compare results with each week's **expected_output.txt**. Week 1 creates sample files inside its own data folder. For all ten commands, see **Course_README.md**.

When you reach Week 10, run **py -3.13 .\Week_10_IR_Project\main.py**. Add **--interactive** for query input and type **quit** to exit.

If setup does not work

Symptom	What to do
py is not recognized	Close and reopen PowerShell. Try <code>python --version</code> . If needed, rerun the installer and enable Add python.exe to PATH.
python opens Microsoft Store	Use <code>py -3.13</code> , or disable Python app execution aliases in Windows Settings.
Missing file or docs folder	Extract the whole ZIP, open PowerShell at the package root, and keep each week's data beside its scripts.
Installer will not run	Check System type. The included executable is Windows x64 only; use the matching python.org installer for other systems.
A lab reports a Python error	Confirm your Python version, then run the exact weekly command from <code>Course_README.md</code> .

Installer source and integrity

The included file is **python-3.13.15-amd64.exe** (29,452,944 bytes). It comes from the official [Python 3.13.15 release page](#) and has a valid Windows digital signature from the Python Software Foundation.

Its SHA-256 matches the checksum published by Python.org:

```
EDEC09C4853AAE9AC36EFB8C9F95B6B8E2FEE65EEE56D9767A8B7C69C574403
```

Optional PowerShell check from the extracted course folder:

```
(Get-FileHash .\Lab_00_Setup\python-3.13.15-amd64.exe -Algorithm SHA256).Hash
```

Windows installation reference: docs.python.org/3.13/using/windows.html